Active antecedent search is guided by pronoun prominence

Carla Spellerberg (cspellerberg@umass.edu), Kyle Johnson, Brian Dillon

Comprehending sentences involves predictive processing about upcoming material, such as where an antecedent to the cataphor in ***His*** *parents thought that* ***John*** *was at school* could be located. Principle C appears to be an active filter in the c-command domain, constraining predictions about upcoming antecedent positions for cataphora [1, 2]. But what is the underlying reason for the apparent effect of Principle C? We compared two hypotheses about the mechanisms underlying antecedent search during the processing of cataphoric pronouns. *Active forward search* predicts that comprehenders make continual predictions about antecedent positions in upcoming material, guided by structural constraints such as Principle C [3]. *Prominence-based search* predicts that forward search is not launched if pronouns are sufficiently prominent (e.g. topical) and that antecedents are inferred instead. In two self-paced reading experiments, we find that pronoun prominence decisively constrains antecedent search, highlighting its key role in pronoun resolution.

Active search is diagnosed using gender mismatch effect (GMME) paradigms [4]: If a name following a cataphor is located in a suitable antecedent position, but mismatches the pronoun in gender (as in ***Her*** *parents thought that* ***John*** *was at school*), a slowdown in reading times can be observed on the name, indicating that comprehenders are recovering from their wrong prediction about the antecedent position. *Active forward search* for antecedents has been shown to be persistent. If no suitable antecedent is found in subject position, the search continues until an antecedent is found later, e.g. in an object position [5].

Predictions: Our two hypotheses can be distinguished by testing for active antecedent search after a structurally inaccessible c-command domain. If Principle C is a selective search filter active in the c-command domain, then the result is that the pronoun is still unresolved when comprehenders exit the c-command domain, *and active search should continue*. Active forward search would therefore predict that comprehenders search for antecedents to cataphors after the c-command domain, which would result in a GMME on the next suitable antecedent after it. Prominence-based search instead predicts that *prominent pronouns do not induce forward search at all*, because an antecedent is immediately inferred; this would result in no search after the c-command domain (or at all), and therefore no GMME on antecedents after the c-command domain.

In **one exploratory** ($N_{subject}$ = 66, N_{item} = 24) **and one (pre-registered) confirmatory SPR experiment** ($N_{subject}$ = 141), we tested our predictions in a 2x2 Latin Square design, crossing the factors *Constraint* and *Match* (see Table 1 for sample stimuli). Each item contained two potential antecedents for the pronoun, one inside and one outside the c-command domain.

Results: We fit Bayesian linear mixed effects models to the two critical and spillover regions in both experiments and to the pooled data from both experiments, following [6] (see model summaries in Table 2/3). The exploratory experiment showed no clear evidence for an interaction of *Match* and *Constraint* in either region, probably due to low power. In the confirmatory and pooled run, we found evidence for an interaction in the predicted direction in the spillover of the first critical region, replicating the standard, c-command sensitive GMME pattern [e.g. 2]. In the second critical region, there was evidence for an interaction under informative priors taken from the posterior of the exploratory experiment in the pooled data, but limited evidence for an interaction under wide and narrow priors. We fit an additional nested model, which showed that the GMME was present in the *No Constraint* conditions, but not in the *Constraint* conditions.

Discussion: We found evidence for an interaction after the c-command domain, such that a GMME was only observed in the *No Constraint* conditions. However, the effect is only visible under informative priors (leading to the expectation that it is small), and was not present in the *Constraint* conditions. Even though more work is needed, the results suggest that the finding

that cataphoric search ‘skips’ the c-command domain arises not because c–command acts as a filter on search, but because prominent pronouns do not engage in forward search at all.

Condition	**Item with matching/mismatching pronoun**
Constraint	***Er/Sie*** entzifferte **Alexanders** Tagebucheinträge zögerlich, aber plötzlich entdeckte ***Philipp*** mittendrin eine Widmung, woraufhin Sarah sehr aufgeregt war.
No Constraint	***Sein/ihr*** Sohn entzifferte ***Alexanders*** Tagebucheinträge zögerlich, aber plötzlich entdeckte ***Philipp*** mittendrin eine Widmung, woraufhin Sarah sehr aufgeregt war.
Translation	‘**He/She/His/Her son** deciphered **Alexander’s** diary entries hesitantly, but suddenly, **Philipp** discovered a dedication in the middle, whereupon Sarah got very excited.’

Table 1: Sample item used in both the exploratory and confirmatory experiment. Cataphoric pronouns and potential referents are bolded. Italics mark available referents according to the predictions of active forward search.

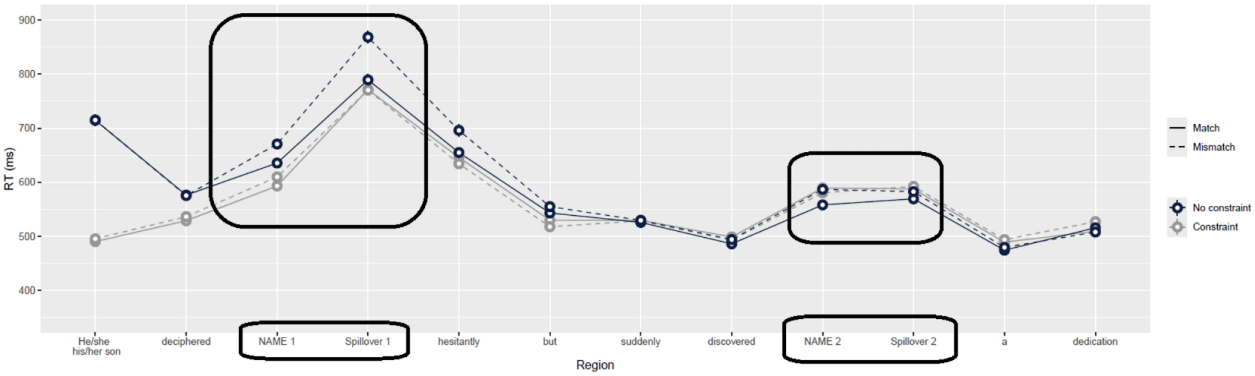

Figure 1: Word-by-word reading times for the pooled data from both experiments. Both critical regions and their spillover regions are highlighted.

Region	**Match**	**Constraint**	**Interaction**
Critical 1	0.02 [-0.02,0.06], *0.02 [-0.02, 0.06]*	-0.08 [-0.12,-0.05], *-0.08 [-0.12, -0.05]*	0.02 [-0.04,0.08], *-0.00 [-0.06, 0.05]*
Spillover 1	0.03 [-0.01,0.07], *0.03 [-0.01, 0.07]*	-0.05 [-0.08,-0.02], *-0.05 [-0.08, -0.02]*	-0.07 [-0.13,-0.01], *-0.07 [-0.12, -0.2]*
Critical 2	0.02 [-0.02,0.05], *0.02 [-0.02,0.05]*	0.04 [0.01,0.08], *0.04 [0.01,0.08]*	-0.03 [-0.10,0.03], *-0.05 [-0.10,0.01]*
Spillover 2	0.01 [-0.02,0.05], *0.01 [-0.03, 0.05]*	0.02 [-0.01,0.05], *0.02 [-0.01, 0.05]*	-0.00 [-0.06,0.05], *-0.02 [-0.07, 0.03]*

Table 2: Posterior estimates and 95% CrIs for the crossed model under wide and informative priors for both critical regions and their spillover regions for E2 (italics mark informative priors).

Region	**Constraint**	**Match: No Constraint**	**Match: Constraint**
Spillover 1	-0.05 [-0.08, -0.02], *-0.05 [-0.08, -0.02]*	0.06 [0.02, 0.11], *0.04 [-0.01, 0.09]*	-0.00 [-0.05, 0.04], *-0.01 [-0.06, 0.03]*
Critical 2	0.04 [0.01, 0.08] *0.04 [0.00, 0.08]*	0.04 [-0.01, 0.08] *0.02 [-0.03, 0.06]*	-0.00 [-0.05, 0.05] *-0.01 [-0.06, 0.03]*

Table 3: Posterior estimates and 95% CrIs for the nested model under wide and informative priors for both the first spillover region and second critical region in E2 (italics mark informative priors).

References: [1] Kazanina, N., E. Lau, M. Lieberman, M. Yoshida, & C. Phillips. (2007). *JML 56*(3), 384-409. [2] Patterson, C. & C. Felser. (2019). *Journal of Psycholinguistic Research 48*(2) 453-475. [3] Chomsky, N. (1981). Dordrecht: Foris. [4] Van Gompel, R.P.G & S.P. Liversedge. (2003). *Journal of Experimental Psychology 29*(1). [5] Giskes, A. & D. Kush. (2021). *Memory & Cognition 49*(7), 1370-1386. [6] Nicenboim, B., S. Vasishth, F. Engelmann & K. Suckow. (2018). *Cognitive Science 42*, 1075-1100.